CDet timing calibration with cross-target run 5710

A start-to-finish account of the procedure, failures, and final closure

CDet group meeting
September 2026

Calibration target and data set

- Cross-target run **5710**
- **2,951,891** available events
- Two CDet layers, each containing 84 half-bars
- 2,688 logical pixel/channel IDs
- ECal provides event selection, position matching, and the timing reference

Goal: align usable pixels and half-bars near 30 ns while removing ECal-time and ToT dependence.

Final observable

Accepted Layer-1/Layer-2 pairs use

$$t_{\text{pair}} = \frac{t_{L1} + t_{L2}}{2}.$$

Its detector-wide width measures the timing performance relevant to paired hits.

The correction model

$$
\begin{aligned}
t_{\text{raw},i} &= 0.01\, t_{\text{TDC},i} - t_{\text{reference}}, \\
t_{\text{pixel},i} &= t_{\text{raw},i} + O_i, \\
t_{\text{ECal},i} &= t_{\text{pixel},i} - (p_0 + p_1 t_{\text{ECal}}) + \Delta_h, \\
t_{\text{final},i} &= t_{\text{ECal},i} - q_L \left(\frac{1}{\sqrt{\text{ToT}}} - \frac{1}{\sqrt{\text{ToT}_{\text{ref},L}}} \right) + s_{\text{run}}.
\end{aligned}
$$

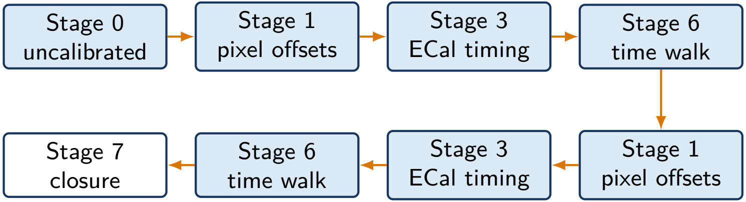
Local corrections

- O_i: one offset per paddle
- Δ_h: later common shift for the 16 pixels in each half-bar

Shared corrections

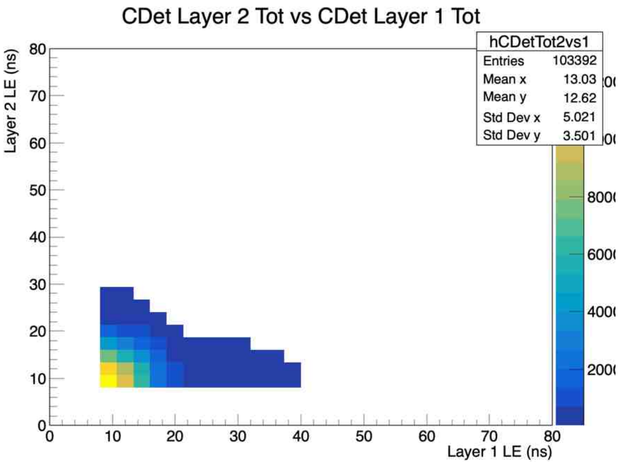
- p_0, p_1: one ECal timing pair for the entire detector
- q_L: one time-walk coefficient per layer
- s_{run}: optional global timing shift for one run, read from `CDet_run<run>.dat`. It is zero for run 5710

Initial two-pass workflow



- Each fit adds a residual correction to the constants already loaded.
- The driver must load the master macro first, or compile the self-contained driver with ACLiC.
- Stage 7 applies pixel offsets, ECal timing, and time walk together.

First correction: the pairing ToT cut



The original pairing fit required

$$8 < \mathrm{ToT} < 40 \text{ ns.}$$

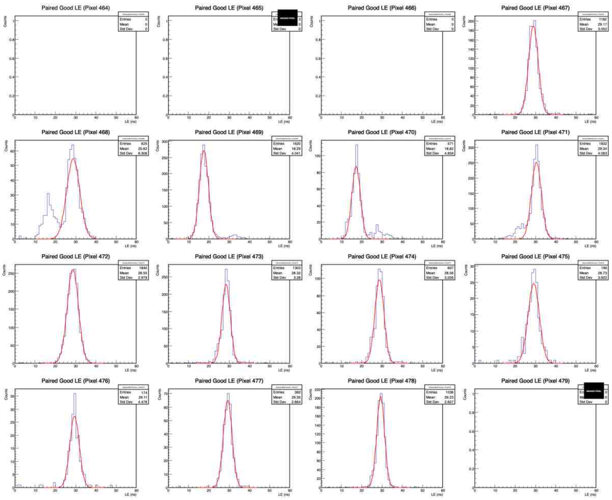
The lower threshold removed a visible population of otherwise useful hits.

We changed the pairing range to

$$\mathbf{4} < \mathrm{ToT} < \mathbf{40} \text{ ns.}$$

Lesson: selection cuts need visual validation at every calibration stage.

The pixel-fit failure was physical, not merely numerical



Bar 29 exposed side peaks from multi-anode PMT cross-talk.

For pixels such as 468, the cross-talk peak could exceed the real peak. A blind fit then selected the wrong population.

Largest peak did not always mean **physical peak**.

Hierarchical pixel-offset extraction

Group estimate

- Divide each 16-pixel half-bar into groups 0–7 and 8–15.
- Sum actual counts without per-pixel normalization.
- Require at least 100 group entries.
- Fit over $-30 < t_{\mathrm{ECal}} - t_{\mathrm{CDet}} < +10$ ns.

Individual estimate

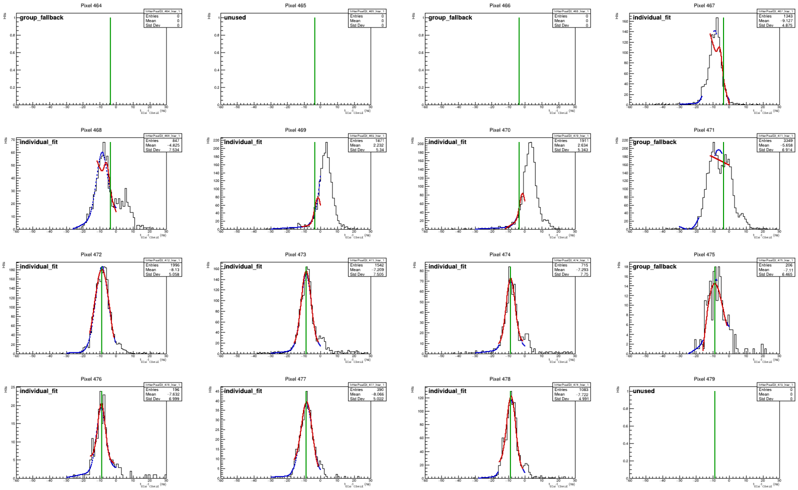
- Require at least 35 pixel entries.
- Seed a narrow fit from the group centroid.
- Accept $0.5 < \sigma < 8$ ns, significance > 1.5, and $\chi^2/\mathrm{NDF} < 15$.
- Use the group centroid as fallback; use a broad individual fit when the group fit fails.

Robust reference

The detector reference is the median of valid individual centroids. The residual update is

$$O_i^{\mathrm{new}} = O_i^{\mathrm{old}} + (\mu_i - \mu_0).$$

Hierarchical diagnostics made every decision inspectable



Group fits, individual fits, fallbacks, and retained offsets are stored together in a hierarchical ROOT file.

Manual LE–ToT polygons resolved ambiguous pixels

The interactive editor displays each flagged pixel in LE versus ToT and saves a `TCutG` around the physical population.

- Candidate ranking deliberately favored sensitivity over specificity.
- We reviewed both banks of all 84 half-bars.
- The final cut file contains **102 manual pixel polygons**.

A manual polygon already resolves the ambiguity. Its 1D timing fit therefore uses an independent seed rather than the group centroid.

Important implementation detail

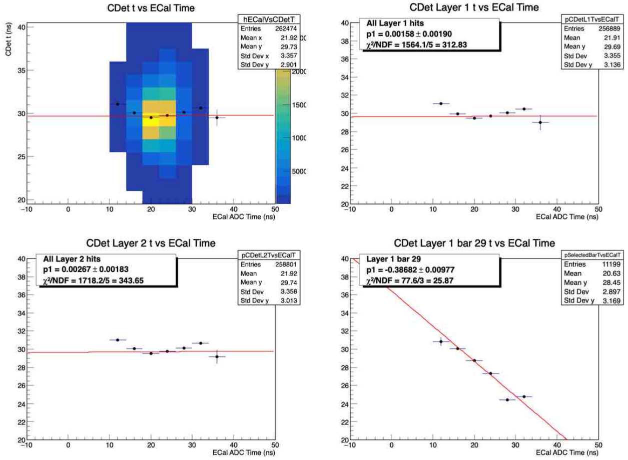
Editing polygons changes only the cut file. The timing constants change only after rerunning

```
extractHierarchicalCDetPixelTimingOffsets(...)
```

Preservation rule

The editor updates cuts in place. Existing cuts remain intact while selected pixels are revisited.

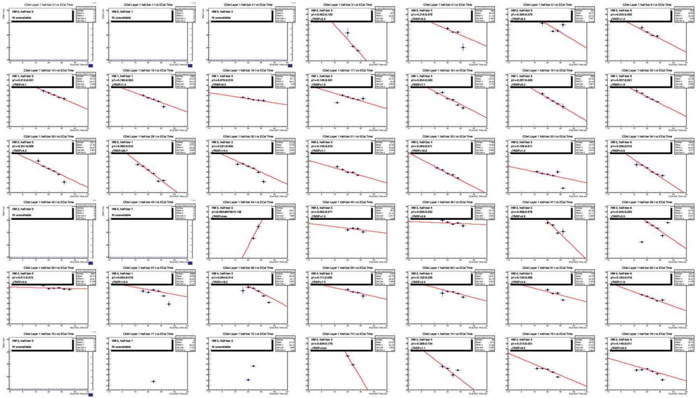
ECal timing: the pooled result looked reassuring



The pooled CDet-versus-ECal profile appeared nearly flat. The selected half-bar showed a strong negative slope.

The aggregate fit concealed the dependence that mattered.

Simpson’s paradox across CDet half-bars



Of 135 fitted half-bars:

- 130 slopes were negative.
- 122 were negative by more than 2σ.
- Mean slope: -0.1284 ns/ns.

Yet the pooled profile remained nearly flat because positive covariance among half-bar means cancelled the negative within-half-bar covariance.

This was a detector-calibration example of Simpson’s paradox.

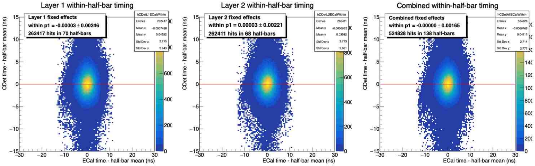
Fixed effects isolated the physical ECal dependence

For hit j in half-bar h, center both variables within that half-bar:

$$x_{hj} = t_{\mathrm{ECal},hj} - \bar{t}_{\mathrm{ECal},h},$$

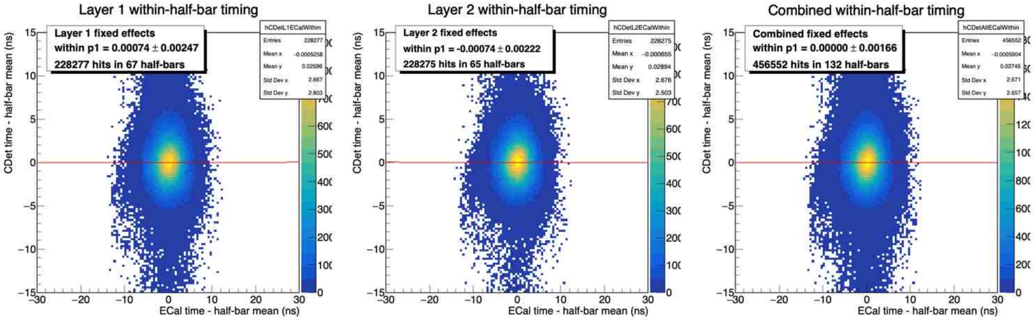
$$y_{hj} = t_{\mathrm{CDet},hj} - \bar{t}_{\mathrm{CDet},h}.$$

Fit one unbinned common slope to (x_{hj}, y_{hj}) while allowing every half-bar its own intercept.



The combined fixed-effects slope, not the pooled profile slope, became the calibration estimator.

ECal-slope closure



Sample	Residual slope (ns/ns)	Statistical uncertainty
Layer 1	+0.000743	0.002474
Layer 2	−0.000738	0.002216
Combined	+0.000000028	0.001660

A flat slope did not guarantee detector-wide alignment

After removing the common ECal slope, half-bars still had different intercepts at the same ECal reference time.

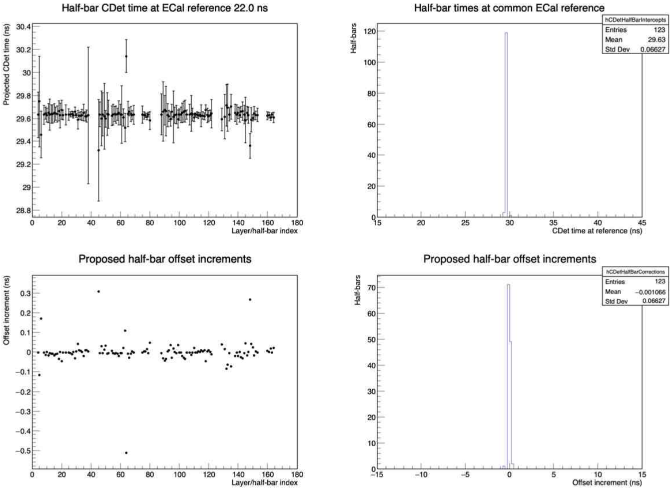
- Reference time: $t_{\mathrm{ECal}} = 22$ ns
- Validated half-bars: 123 of 168
- Detector reference: 29.6304 ns
- Between-half-bar RMS before alignment: 1.61545 ns

Each accepted half-bar received one common additive increment across its 16 pixels:

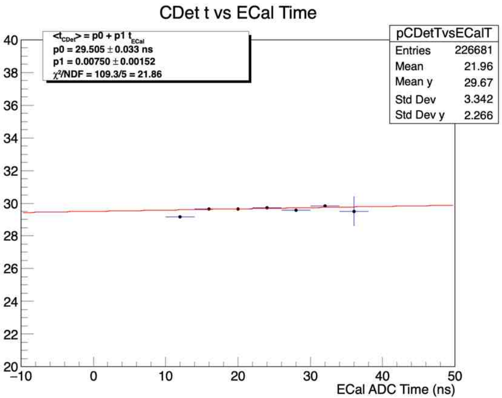
$$\delta O_h = t_{\mathrm{reference}} - t_h(22\ \mathrm{ns}).$$

This preserves the within-half-bar ECal slope and the relative pixel structure.

Half-bar intercept closure



Final calibrated timing



Final constants

$$p_0 = 74.192800 \text{ ns}, \qquad p_1 = 0.817261.$$

- Between-half-bar RMS: **0.03961 ns**
- Individual-hit RMS: **2.99623 ns**
- Paired-layer mean-time width: ≈ 2.27 **ns**
- Predicted gain from another alignment: only 0.00026 ns

The large change in p_0 relative to the older calibration agrees qualitatively with the ECal group's roughly 100 ns database timing shift.

What we learned

- Cross-talk can make the largest 1D timing peak the wrong physical population. LE–ToT information resolves that ambiguity.
- Group fits help seed sparse pixels, but real counts must remain unnormalized and group-fit χ^2 should remain diagnostic.
- Manual polygons should bypass the group-centroid constraint because the user has already identified the physical population.
- A pooled timing slope can hide strong within-detector trends. Half-bar fixed effects provide the relevant ECal correction.
- Removing a common slope and aligning half-bar intercepts solve different problems. Both steps matter for detector-wide resolution.
- Every automated update needs a diagnostic-only mode, preserved inputs, auditable fit results, and a final closure run.

Definitive run-5710 products

The final archive is

`CDet_run5710_halfbar_aligned_final_archive/`.

It contains:

- the definitive calibration file;
- the 102 manual LE–ToT polygon cuts;
- hierarchical pixel-fit results and ROOT diagnostics;
- applied half-bar increments and final closure records;
- final pooled, fixed-effects, and intercept-alignment figures;
- SHA-256 checksums for the primary artifacts.

Repository checkpoint

Tag: `cdet-run5710-halfbar-aligned-final-v2.0`

Appendix: full calibration command

```
.L PlotElastic_Calibration_Master_stageflag_singlefile_crosstarget.C+
.L Run_CDet_Calibration_TwoPass_InSession_AllCross_Individually.C

Run_CDet_Calibration_TwoPass_InSession_AllCross_Individually(
    5710, -1, 0, -1, -1,
    0.02, 60, 4, 50,
    1, 100, 1, 100,
    0.10, 0.02, 0.1,
    3, false, 30, 2, 1, true);
```

The final `true` removes the active calibration and starts fresh. Back up `CDet_calibration_dt.dat` before using it.

Appendix: validation and resolution

```
.L PlotElastic_Calibration_Master_stageflag_singlefile_crosstarget.C+

PlotElastic_Calibration_Master_stageflag_singlefile_crosstarget(
    "CDet_run5710_event_display.conf");

plotCDetLayersTimeComp("CDet_run5710_event_display.conf");

reportCDetPairedTimeResolution();
```

The resolution helper reports the RMS of the full accepted-pair mean-time distribution separately from a Gaussian fit to its central 25–35 ns core.